

GYZYLARBAT, Turkmenistan

WMO: 387630

Lat: 38.983N

Lon: 56.283E

Elev: 302

StdP: 14.54

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.4	21.2	6.6	7.8	18.6	12.8	10.6	29.6	31.7	35.9	27.5	41.2	9.8	90	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	105.8	71.3	103.3	70.8	101.0	70.5	78.8	94.9	76.9	94.1	75.0	93.2	8.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	130.1	89.1	71.3	116.7	87.8	68.9	107.5	88.0	42.6	94.2	40.5	93.9	38.8	93.2	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	23.3	21.0	12.3	110.8	10.0	3.0	5.2	112.9	-0.7	114.7	-6.3	116.4	-13.5	118.6
			11.2	82.0	9.8	3.1	4.1	84.2	-1.6	86.1	-7.1	87.8	-14.2	90.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	37.7	40.7	51.5	63.5	76.1	85.3	89.8	87.9	77.0	63.1	49.1	39.9	(6)
	DBStd	20.13	8.67	9.50	8.08	8.39	6.40	5.60	4.57	4.96	6.41	8.25	8.89	7.77	(7)
	HDD50	1192	387	274	76	5	0	0	0	0	0	9	116	325	(8)
	HDD65	3478	846	680	426	124	5	0	0	0	4	133	481	779	(9)
	CDD50	6152	6	14	122	410	809	1059	1232	1176	809	415	89	11	(10)
	CDD65	2963	0	0	7	79	349	609	767	711	363	74	4	0	(11)
	CDH74	39940	0	1	109	782	3707	8459	11693	10505	4011	641	32	0	(12)
	CDH80	23223	0	0	30	284	1717	5006	7492	6607	1899	180	8	0	(13)
(14) Wind	WSAvg	8.6	8.4	9.1	9.7	9.9	8.5	9.1	8.2	7.8	8.5	8.0	7.9	8.5	(14)
(15) Precipitation	PrecAvg	8.0	0.9	1.0	1.0	0.9	0.9	0.3	0.2	0.1	0.2	0.6	0.7	0.9	(15)
	PrecMax	13.9	2.9	2.4	2.8	3.3	4.5	2.8	2.6	0.8	1.8	2.4	2.6	2.6	(16)
	PrecMin	3.4	0.0	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.6	0.6	0.6	0.6	0.7	0.9	0.6	0.5	0.2	0.4	0.6	0.6	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	69.9	84.2	93.5	100.5	106.8	109.7	109.3	100.8	90.2	78.3	66.2	(19)
		MCWB	49.4	51.6	58.6	65.5	69.1	71.2	73.5	72.6	69.6	65.4	58.5	50.4	(20)
	2%	DB	57.5	62.2	76.0	87.4	95.8	103.5	106.1	105.3	96.1	84.7	69.6	59.6	(21)
		MCWB	47.3	50.0	56.7	64.1	67.2	70.1	71.9	71.0	68.1	64.2	55.8	49.3	(22)
	5%	DB	52.3	56.9	69.8	82.6	92.4	100.9	103.6	102.7	93.0	80.8	65.5	54.8	(23)
		MCWB	45.4	48.0	55.1	61.8	66.5	69.6	71.5	70.6	67.4	62.0	54.1	47.4	(24)
	10%	DB	48.5	53.0	64.8	78.0	89.1	98.1	101.2	100.0	89.8	77.1	61.7	50.7	(25)
		MCWB	43.2	46.0	52.7	59.9	65.9	69.1	71.3	70.3	66.5	60.3	52.6	45.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	55.4	61.0	68.1	73.5	78.3	81.7	81.6	75.7	70.0	61.2	53.6	(27)
		MCDB	61.0	66.3	76.4	87.3	90.6	96.5	96.8	96.3	91.6	82.9	72.3	63.2	(28)
	2%	WB	48.1	50.9	57.9	65.4	70.7	75.4	79.1	78.7	73.0	66.6	58.1	50.2	(29)
		MCDB	56.4	59.4	71.9	83.4	89.3	94.2	95.5	93.4	89.5	80.2	66.3	57.8	(30)
	5%	WB	45.7	48.5	55.7	63.3	68.9	73.7	77.4	76.5	70.9	64.1	55.5	47.9	(31)
		MCDB	51.8	56.0	68.1	79.3	87.0	92.3	94.8	93.7	87.4	77.5	63.9	54.1	(32)
	10%	WB	43.5	46.3	53.5	61.3	67.3	72.2	75.6	74.4	68.6	61.8	53.3	45.6	(33)
		MCDB	48.1	53.0	63.4	75.3	84.6	91.0	93.4	93.1	86.0	74.9	60.7	50.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	14.9	17.9	19.1	20.2	20.8	20.1	21.9	22.4	20.9	15.5	12.3	(35)
		MCDBR	20.7	22.7	27.3	26.5	25.2	24.9	23.3	25.1	26.2	25.0	21.5	20.2	(36)
	5% WB	MCWBR	12.3	13.0	12.4	10.8	9.9	9.7	9.3	10.3	12.7	12.3	11.1	12.5	(37)
		MCWBR	18.8	20.3	23.7	23.3	21.7	19.8	18.7	19.0	20.6	21.3	17.8	18.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.403	0.453	0.509	0.575	0.560	0.559	0.578	0.530	0.505	0.500	0.439	0.409	(40)
		taud	2.049	1.920	1.816	1.720	1.784	1.795	1.760	1.847	1.877	1.869	1.995	2.034	(41)
	Ebn at Noon		230	234	235	229	236	236	230	238	233	215	213	215	(42)
		Edh at Noon	39	50	62	72	69	69	70	63	57	52	41	37	(43)
(44) All-Sky Solar Radiation	RadAvg		668	927	1307	1676	2113	2437	2335	2171	1779	1235	824	606	(44)
	RadStd		60	103	89	167	104	69	89	105	62	70	62	67	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page